

Cybersecurity: Social Engineering Attacks

Social engineering, **physical security** and **data controls** are considered **pervasive**—since these three types of cybersecurity elements are applied universally across *IT security*.

Please provide a **short concise** definition/explanation for each of the cybersecurity terminologies. Each definition/explanation entry is worth **5** points.

Direction & Submission: use the downloaded document, provide the definition/explanation for each indicated row below. At the completion, submit the completed document through Canvas. You have up to **one weeks** to submit the assignment after the due date. Please note that there is a daily 5% deduction if submitted after the due date.

	Terms	Definition/explanation
1	Social Engineering	Social engineering in computer security is the manipulation of individuals into divulging confidential information, such as passwords or personal details, through deception or psychological tactics.
2	Human vector	refers to the use of human psychology as the primary attack surface to compromise security systems.

3	Human manipulation for social engineering (provide definition and example)	a. Authority: Authority in social engineering is a tactic where attackers exploit the perceived power or credibility of a figure, such as a boss, IT administrator, or law enforcement officer, to manipulate victims into compliance. A hacker impersonates a company's CEO and sends an urgent email to the finance department requesting a wire transfer. b. Intimidation: Intimidation in social engineering is a tactic where attackers use fear, threats, or pressure to manipulate victims into complying with their demands. A scammer posing as an IRS agent calls a victim, claiming they owe back taxes and will be arrested if they don't make an immediate payment. Panicked, the victim complies and transfers money to the attacker. c. Consensus: is a tactic where attackers manipulate victims by making them believe that others have already complied with a request, making it seem more legitimate and trustworthy. A scammer sends an email to employees claiming that multiple coworkers have already completed a "mandatory security update" by clicking a provided link. d. Scarcity: is a tactic where attackers create a sense of urgency or limited availability to pressure victims into acting quickly without verifying legitimacy. A phishing email claims that a user's bank account will be permanently locked in the next 30 minutes unless they immediately verify their credentials through a provided link. Panicked by the urgency, the victim clicks the link and unknowingly enters their login details into a fraudulent site. e. Urgency: is a tactic where attackers create a time-sensitive situation to pressure victims into making quick decisions without proper verification. f. Familiarity: is a tactic where attackers build trust by pretending to have a connection with the victim. g. Trust:
Type social Engineering		

4	Phishing	is a social engineering attack where an attacker impersonates a trusted entity to deceive individuals into revealing sensitive information, such as passwords, credit card numbers, or personal details.
5	• Spear phishing	<i>is a targeted form of phishing where attackers customize their fraudulent messages to a specific individual or organization</i>
6	• whaling	is a type of phishing attack specifically aimed at high-profile targets, such as executives or important figures within an organization, with the goal of stealing sensitive information or gaining access to critical systems.
7	• vishing	is a type of social engineering attack where attackers use phone calls or voice messages to impersonate legitimate organizations or individuals
8	• smishing	is a type of phishing attack that uses text messages (SMS) to deceive individuals into revealing personal information
9	Impersonation	when an attacker pretends to be someone else, such as a trusted individual or authority figure, to gain access to confidential information, systems, or services.
10	• Brand impersonation	type of social engineering attack where an attacker impersonates a well-known brand or company to deceive individuals
11	• Business email compromise (BEC)	is a type of cyberattack where an attacker gains access to a business's email system and uses it to impersonate executives or employees.
12	• Domain spoofing	is a type of cyberattack where an attacker forges the sender's email address or website domain to make it appear as though the communication is coming from a trusted source.
13	• Identity theft	is the deliberate use of someone else's personal information, such as Social Security numbers, bank account details, or credit card information, without permission.
14	Redirection	where an attacker manipulates a victim's web traffic by redirecting them from a legitimate website to a malicious or fraudulent site
15	Typo squatting	type of domain name spoofing where an attacker registers a domain name that is a common misspelling or typo of a popular website.

16	misinformation	refers to the deliberate spread of false or misleading information to manipulate public opinion, create confusion, or deceive individuals.
17	disinformation	is the intentional spread of false or misleading information with the goal of deceiving others, often for political, financial, or strategic purposes.
18	hoax	a deliberate attempt to deceive or trick people into believing a false claim or story, often for entertainment, to cause panic, or to manipulate behaviors.
19	Watering hole attack	the attacker compromises a specific website that is frequently visited by a particular group of individuals or organization.
20	Data reconnaissance	is the process of gathering information about a target system, organization, or individual to identify vulnerabilities or valuable data.
21	• Dumpster Diving	is the practice of searching through trash or discarded materials to find valuable information
22	• Google docking	refers to the practice of using Google's search engine to gather detailed information about a target
23	• Shoulder surfing	is a type of physical security breach where an attacker observes someone's activities, such as typing passwords, PINs, or other sensitive information, by looking over their shoulder in public spaces or crowded environments.
Physical security controls		
	Perimeter Defense	
24	Barriers	in cybersecurity are protective measures, such as encryption, firewalls, and security training, designed to prevent unauthorized access and manipulation.
25	• Fencing	in cybersecurity refers to the use of physical or digital boundaries, such as firewalls, network segmentation, or access controls, to restrict unauthorized access to systems or data.
26	• lighting	in security refers to the strategic use of illumination to deter unauthorized access, enhance surveillance, and improve the safety of physical spaces.
27	• A bollard	is a sturdy, short post used to control or restrict vehicle access and enhance physical security in areas like buildings, sidewalks, or pedestrian zones.

28	Security guards	are trained personnel responsible for protecting people, property, and information by monitoring, patrolling, and responding to security threats.
29	Video surveillance	is the use of cameras to monitor, record, and analyze activities for security, safety, and crime prevention purposes.
30	Closed-circuit television (CCTV)	is a private video surveillance system that uses cameras to transmit footage to a limited set of monitors for security and monitoring purposes.
31	Unmanned aerial vehicles	are remotely operated or autonomous aircraft used for surveillance, data collection, and security monitoring without an onboard human pilot.
32	sensors	are devices that detect and respond to physical inputs such as motion, temperature, light, or sound to enhance security, automation, and monitoring systems
33	infrared	refers to electromagnetic radiation with wavelengths longer than visible light, often used in security systems for night vision, motion detection, and heat signature detection.
42	Microwave sensor	use microwave signals to detect movement or changes in the environment, commonly employed in motion detection systems for security purposes.
34	Ultrasonic sensors	use high-frequency sound waves to detect objects, measure distances, or monitor movements, often used in security systems and automation.
35	ultrasound	refers to sound waves with frequencies higher than the human hearing range, commonly used in imaging, monitoring, and distance measuring technologies, including in security systems.
36	Pressure sensor	detect changes in force or pressure applied to a surface or material, often used in security systems to monitor doors, windows, or other entry points for unauthorized access.
37	A security buffer	is a protective zone or additional layer in a system or process designed to prevent unauthorized access or attacks by providing extra time, space, or resources to respond to threats.
38	Access badge	physical card or device that grants authorized individuals entry to restricted areas
39	mantrap	is a small, controlled entryway with two doors that are interlocked, allowing only one door to open at a time, used to prevent unauthorized access or to verify identity before entry.

40	Access control vestibule	is a secured entry area with two doors, where one door remains locked until the previous door is closed, used to enhance security by controlling and verifying access to restricted areas.
41	Reception area (<i>medium security</i>)	is a controlled entry point in a building where visitors are greeted, verified, and monitored before being granted access to restricted areas,
42	Waiting room (<i>low security</i>)	is a designated area where visitors can wait, typically without stringent security measures, before being granted access to other parts of a building or facility.
43	Electronic lock	is a locking mechanism that uses electronic components, such as keypads, cards, or biometric data, to control access,
	Preventing Data Leakage (<i>Physical control</i>)	
44	Faraday Cage	is an enclosure made of conductive materials that blocks external electric fields, electromagnetic radiation, or signals, providing protection for sensitive electronics or securing areas from electronic surveillance.
	Faraday bags	are portable, electromagnetic shielding pouches designed to block radio frequency signals, protecting electronic devices from hacking, tracking, or remote data access.
45	Electromagnetic interference (EMI)	is the disruption of electronic devices or systems caused by external electromagnetic fields
	Protected Distributive System	
46	Protected distributive system (PDS)	is a secure communication framework designed to ensure the confidentiality, integrity, and availability of data transmitted across distributed networks
47	Hardened carrier PDS	refers to a protected distributive system that is specifically designed with additional security measures, such as encryption and physical shielding
48	Alarmed carrier PDS	is a protected distributive system that incorporates alarm mechanisms to detect unauthorized access or security breaches
	Computer hardware security	
49	Cable lock	is a flexible security device typically made of steel or strong materials
50	Data Controls	
	Data classifications	refer to the process of categorizing data based on its sensitivity, value, and importance, determining

		the level of protection and access controls required for each category.
51	Confidential	data is sensitive information that is intended to be kept private and is protected by access restrictions to prevent unauthorized disclosure or misuse.
52	Private data	refers to personal or sensitive information that is intended to be kept secret or limited to specific individuals or organizations, typically requiring consent for access or sharing.
53	Sensitive data	refers to information that requires heightened protection due to its potential to harm individuals or organizations if disclosed
54	critical	refers to information essential to the operation, security, or survival of an organization
55	public data	refers to information that is available to the general public and does not require any special access controls or restrictions
58	restricted	information that is limited in access due to its sensitivity or importance
59	Data/information classification levels	are categories used to define the sensitivity and access requirements of data
60	Top secret	is the highest classification level for sensitive information
61	secret	is a classification level for sensitive information
62	confidential	is a classification level for information that, if disclosed without authorization, could cause harm or damage to an organization or individual
63	Sensitive but unclassified (SBU)	refers to information that is not classified but still requires protection due to its sensitive nature
	Type of Data	
64	regulated	refers to information that is subject to specific legal, regulatory, or industry requirements for protection
65	Intellectual property (IP)	refers to creations of the mind, such as inventions, designs, trademarks, and copyrights, that are legally protected from unauthorized use or reproduction.

66	Trade secret	confidential business information, such as formulas, processes, or methods, that provide a competitive advantage
67	Enterprise information	data and knowledge generated, used, or stored by an organization, which is critical for decision-making, operations, and business processes.
79	Legal information	data related to laws, regulations, legal proceedings, contracts, or other legal matters
68	Financial information	data related to an organization's financial status, transactions, budgets, and assets
69	Human-readable	data or information formatted in a way that is easily understood by humans, without requiring specialized knowledge or tools to interpret.
70	non-human-readable (<i>machine readable</i>)	data or information formatted in a way that can be processed and understood by computers or machines, but not easily interpreted by humans without specialized tools.
	Data breach consequences	
71	Reputation damage	refers to harm caused to an individual's or organization's public image, credibility, or trustworthiness
72	IP theft	refers to the unauthorized use, reproduction, or distribution of intellectual property
73	Fines	are monetary penalties imposed by a legal or regulatory authority as punishment for violating laws, regulations, or contractual obligations.
	Protecting Data	
74	State of data	current condition or status of data, which can include its storage, processing, transmission, or archival, and can be classified into different states like data at rest, data in transit, or data in use.
75	Data in processing (in use)	data that is actively being accessed, manipulated, or analyzed by a computer or application
76	Data in transit	data that is actively being transmitted over a network from one location to another
77	Data at rest	data that is stored on physical media, such as hard drives or databases

78	Geolocation (data in moving device)	tracking and collection of geographic location data from a device, such as a smartphone or GPS-enabled device, as it moves or changes position over time.
	Data security methods	
79	Data minimization	is the principle of collecting only the necessary amount of data required for a specific purpose
80	Data masking	is the process of obscuring or replacing sensitive data with fictitious or scrambled values to protect it during testing, development, or other non-production uses while maintaining its usability.
81	Tokenization	the process of converting sensitive data, such as credit card numbers, into a unique identifier or token that retains essential information for processing but is meaningless outside of the secure system.
82	Restriction	refers to the imposition of limits or controls on access, usage, or distribution of data, information, or resources to protect confidentiality, integrity, or security.